Feature Analysis & Model Explanation

VayuBudhi Air Quality Models

Part 1: Feature Analysis (What data we look at)

We pulled real-world data from open-source weather and air quality APIs. The core parameters (variables) we feed into the model are:

Pollutants: pm25 (fine smoke/exhaust particles) and pm10 (larger dust particles).

Basic Weather: temp (temperature in Celsius) and humidity (relative humidity %).

Wind & Atmosphere: wind_speed (how fast the air is moving) and pblh (Planetary Boundary Layer Height).

The 'Formulas' (Feature Engineering)
Instead of just throwing raw numbers at the AI, we combined them into physics-based formulas so the AI understands why pollution happens.

1. The Ventilation Index (The 'Pollution Trap' Formula)

Formula: Ventilation Index = pblh * wind_speed

In simple words: The pblh is like the 'ceiling' of the sky. If the ceiling is very low and there is no wind, pollution gets completely trapped near the ground like a lid on a pot. A low Ventilation Index tells the model: 'Expect a massive pollution spike soon.'

2. The Particle Ratio (The 'Dust Finder' Formula)

Formula: PM Ratio = pm25 / pm10

In simple words: PM10 includes giant dust particles, while PM2.5 is mostly fine smoke and vehicle exhaust. If this ratio is very low (meaning there is way more PM10 than PM2.5), it almost certainly means a Dust Storm or construction dust, rather than tailpipe exhaust.

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Part 2: Building the Models

We built two completely separate machine learning models to solve two different problems.

Model 1: Source Attribution (Who is causing the pollution?)

The Model Type: RandomForestClassifier (A decision-tree model that votes on categories).

How it was built:
Because we don't have human inspectors telling us the exact source of pollution for every historical data point, we used a technique called Weak Supervision Heuristic Labeling. We wrote a few simple 'if/then' rules (heuristics) to auto-label the data, and then taught the model to learn the deeper patterns:

- Rule 1: If PM Ratio < 0.35 -> Label as 'dust'

- Rule 2: If it is Rush Hour (7-10 AM or 5-8 PM) -> Label as 'vehicular'

- Rule 3: If it is mid-day (11 AM - 4 PM) -> Label as 'industrial'

- Rule 4: Otherwise -> Label as 'biomass_burning' (trash/crop fires).

The Magic (Conformal Prediction): We wrapped this model in a math library called MAPIE. Instead of the model just blindly guessing 'Vehicular,' the MAPIE formula forces it to be honest about its uncertainty. It says: 'I am 90% confident the source is either [Vehicular OR Industrial].' If the evidence is weak, it outputs a bigger list so you know to send a drone to verify first!

Model 2: The AQI Forecaster (What will the air be like tomorrow?)

The Model Type: XGBoostRegressor (A powerful algorithm that predicts exact continuous numbers).

How it was built:
- The Target (What it predicts): We wrote a script that looks at the pm25 at a specific station, and then looks exactly 24 hours into the future to find the target_y_reg (tomorrow's PM2.5).
- We fed the model the current weather, current pollution, and the Ventilation Index. XGBoost is exceptionally good at finding non-linear patterns (for example, it learns that if the Temperature drops and the Ventilation Index drops, PM2.5 will exponentially multiply overnight).

The Magic (Conformal Prediction): Just like the classifier, we used MAPIE for the forecaster. Instead of saying 'Tomorrow's AQI will be exactly 154', the math formula creates a calibrated bracket: 'Tomorrow's AQI will be 154, but based on the chaotic wind data, I am 90% sure it will land somewhere strictly between 130 and 170.'